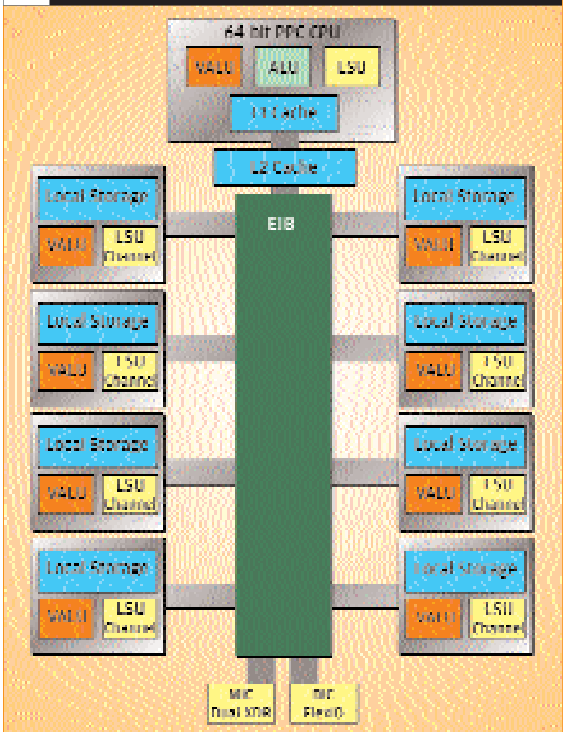
The Cell Architecture



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you call in three lesser geniuses with an IQ of 140 each. Obviously, three heads are better than one, so you take the second route. This is exactly what the people developing the Cell did. They put in multiple processors so various threads (streams of instructions) can be processed simultaneously.

So how did they manage to make a 'simpler' processor much faster than its more complex brethren? Well, surprisingly, they did it by making the processor *cores* simpler. The Cell processor has only a two-issue core, meaning, it can process only two instructions at a time, compared with the Athlon 64s, which are three-issue cores. You can see that the Cell is a simplified processor, with slightly limited capabilities.

Now, why make something simpler when millions of dollars were pumped into making processor cores complex in the first place?

The answer, in a word, is performance.

Over the past 20 years, every processor evolution has managed to add more and more transistors to increase performance. Now all instructions have various offshoots, as in everyday life you may make decisions based on things such as "if X happens, we'll go with Y, but if Z happens, we'll go with A", and so on.

The processor makes the same kind of decisions and attempts to guess which is the right way to go. Guessing wrong is costly in terms of wasted instruction cycles, and, increases the time taken to 'process' instructions.

To prevent this wrong guessing, or to at least minimise it's impact, processor dies are crammed with branch predictors, caches, and predictive cache pre-fetch units. This has no doubt massively improved performance, but it has also led to much higher power consumption and hence excessive heating along with an impossibly large number of transistors on the processor chip.

A few years ago, the benchmark software and applications were office suites and photo editing software, which drove the development of processors that were optimised for single-thread (single application, single task) operations.

The scenario now is vastly different with 3D applications and gaming being the performance benchmarks. More and more manufactures are thinking of taking computers and the processing power to grow beyond the purview of just games and applications and enter the domain of technology as an enabler.

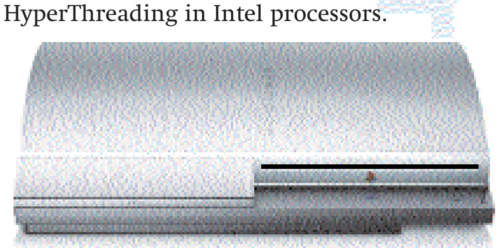
At the latest Intel Developer's Forum (IDF), things such as real-time voice recognition, unstructured search, and life-like in-game physics and AI (artificial Intelligence) have prompted manufacturers to look beyond single-thread optimisation, and focus on parallel computing where the PC either has multiple processors or has a multi-core processor that can process multiple threads simultaneously.

This approach is also important from a long-term point of view. Considering the way in which the demand for processing power has increased in the last five years, there is no telling the amount of power we would need five years from now.

This above-mentioned idea of increasing parallel processing has been applied in the creation of the cell processor. What follows is an overview of how the cell processor works.

The PPE

The cell processor has a 64-bit PowerPC chip, which has been termed as the PowerPC Processing Element, or the PPE. The PPE is a general-purpose CPU, which means that anything you can do with your Athlon 64 or Intel processors, you can do with the PPE. It features 64 KB of L1 cache and 512 KB of L2 cache. It also incorporates something called 'Simultaneous Multi Threading', or SMT, which is similar to HyperThreading in Intel processors.



Future of the gaming console: the PS3

In terms of head-to-head performance, there is nothing too special about the PPE, and pitched against Intel EE and Athlon FX processors, it would surely lose. But what the PPE signifies is a shift in developers' attitude—of moving away from complex, single-thread-optimised processors to simpler processors working in parallel.

Apart from the PPE, the cell has eight Synergistic Processing Elements (SPEs). These are similar to the simpler processors of a few years ago, except they've been designed from the ground up with specific functions in mind.